INCH-POUND
MIL-PRF-3098/102C
May 17, 2004
SUPERSEDING
MIL-PRF-3098/102B
1 October 1997

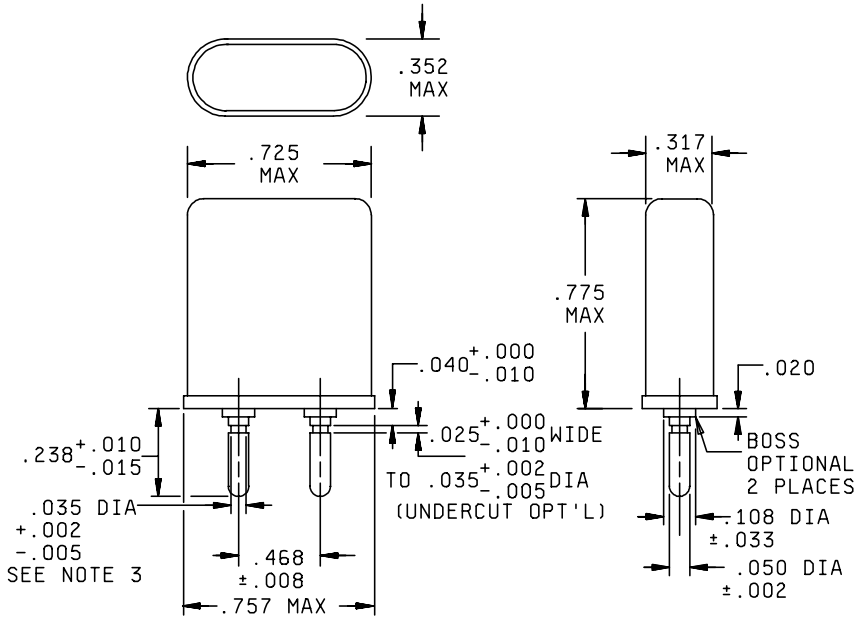
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR125/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 1.85 MHz to 3.05 MHz; fundamental mode; noncontrolled; antiresonance.




| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .002 | .05 | .020 | .51 | .050 | 1.27 | .486 | 12.34 |
| .005 | .13 | .025 | .64 | .108 | 2.74 | .725 | 18.42 |
| .008 | .20 | .033 | .84 | .238 | 6.05 | .757 | 19.23 |
| .010 | .25 | .035 | .89 | .317 | 8.05 | .775 | 19.69 |
| .015 | .38 | .040 | 1.02 | .352 | 8.94 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR125/U.



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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 1.85 MHz to 3.05 MHz, inclusive.

Capacitance, shunt: 7 pF, maximum.

Frequency tolerance: ±0.005 percent.

Mode of oscillation: Fundamental.

Antiresonance, load capacitance: 32.0 pF ±.05 pF.

Rated drive level: 1.0 mW, maximum.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Equivalent resistance: See table I.

TABLE I. Equivalent resistance.

| Frequency Range Inclusive | Maximum Resistance |
|---|---|
| MHz | Ohms |
| 1.85 to 1.999999 | 550 |
| 2.0 to 2.249999 | 500 |
| 2.25 to 2.999999 | 320 |
| 3.0 to 3.050000 | 175 |

| Shock: | Below 2 MHz | 2 MHz and above |
|---|---|---|
| Frequency change permitted: | ±0.001 percent | ±0.0005 percent |

Equivalent resistance change permitted: ±15 percent or ±2 ohms, whichever if greater.

Vibration: Method 204, MIL-STD-202, test condition B; amplitude to produce 5 gravity units, maximum.

| | Below 2 MHz | 2 MHz and above |
|---|---|---|
| Frequency change permitted: | ±0.001 percent | ±0.0005 percent |

Equivalent resistance change permitted: ±15 percent or ±2 ohms, whichever if greater.

Temperature cycling: ±0.001 percent.

Aging:

Frequency change permitted: ±0.002 percent.

Equivalent resistance: See table I.

Aging temperature: 105°C for 200 days.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5955-0756)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil.



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